Practical No. 3

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#Python
#Queue Data Structure - Call Center
queue = []
def addcall(customer_id, call_time):
    call = (customer_id, call_time)
    queue.append(call)
    print("Call added successfully.")
def answercall():
    if len(queue) == 0:
        print("No calls waiting.")
    else:
        call = queue.pop(0)
        print("Answered Call:", call)
def viewqueue():
    if len(queue) == 0:
        print("Queue is empty.")
    else:
        print("Waiting Calls:")    for call in queue:       print(call)
def isqueueempty():
    return len(queue) == 0
while True:
    print("\n--- Call Center Menu ---")
    print("1. Add Call")
    print("2. Answer Call")
    print("3. View Queue")
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    print("4. Check Queue Empty")
    print("5. Exit")
  choice = input("Enter your choice: ")

if choice == '1':
   customer_id = input("Enter Customer ID: ")
   call_time = input("Enter Call Time: ")
   addcall(customer_id, call_time)

elif choice == '2':
   answercall()

elif choice == '3':
   viewqueue()

elif choice == '4':
   if isqueueempty():
      print("Queue is empty.")
   else:
      print("Queue is not empty.")

elif choice == '5':
   print("Exiting Call Center System.")
   break

else:
   print("Invalid choice. Try again.")
```

Sample Output :

--- Call Center Menu ---

1. Add Call
2. Answer Call
3. View Queue
4. Check Queue Empty
5. Exit Enter your choice: 1

Enter Customer ID: C101
Enter Call Time: 10:00 AM
Call added successfully.

--- Call Center Menu ---

1. Add Call
2. Answer Call
3. View Queue
4. Check Queue Empty
5. Exit Enter your choice: 1

Enter Customer ID: C102
Enter Call Time: 10:05 AM
Call added successfully.

--- Call Center Menu ---

1. Add Call
2. Answer Call
3. View Queue
4. Check Queue Empty
5. Exit Enter your choice: 3

Waiting Calls:
('C101', '10:00 AM')
('C102', '10:05 AM')

--- Call Center Menu ---

1. Add Call
2. Answer Call
3. View Queue
4. Check Queue Empty
5. Exit

Enter your choice: 2
Answered Call:
('C101', '10:00 AM')

--- Call Center Menu ---

1. Add Call
2. Answer Call
3. View Queue

4. Check Queue Empty
5. Exit

   Enter your choice: 4

   Queue is not empty.

--- Call Center Menu ---

1. Add Call
2. Answer Call
3. View Queue
4. Check Queue Empty
5. Exit

   Enter your choice: 5

   Exiting Call Center System.